

Asmith. Pampana

CREATIVE
TECHNOLOGIST

LOCATION

Hyderabad, India

EMAIL

pampana.asmith@gmail.com

PORTFOLIO

asmithcodes.github.io

GITHUB

github.com/Asmithcodes

LINKEDIN

in/asmithpampana

EXPERTISE

SOFTWARE & WEB

JavaScript / TS

Python

React

Node.js

MERN Stack

HTML/CSS

HARDWARE & SYSTEMS

C/C++

ESP32 / Arduino

IoT Integration

Embedded Systems

DATABASES

MongoDB

PostgreSQL

OracleSQL

Final-year ECE student and software developer specializing in functional web applications and system prototypes. Passionate about rapid iteration, bridging hardware and software, and building reliable, user-centric products with modern technologies.

Selected Projects

Lumina - AI Decision Engine

Web Application

GitHub

Demo

- Engineered an AI-powered decision analysis tool using the Google Gemini API.
- Designed intelligent algorithms to surface hidden assumptions and blind spots.

Acoustic Data Link

Signal Processing / React

GitHub

Demo

- Built a browser-based 8-FSK acoustic data link for air-gapped P2P data transfer.
- Designed SNR handshake detection and CRC32 verification, achieving <1% error rate.

QR A3 Batch Generator

React / TypeScript / Vite

GitHub

Demo

- Built a lightweight web app to generate bulk QR codes arranged on A3 print sheets.
- Implemented layout customization and high-resolution export workflows.

Wireless Head Tracking System

C++ / ESP32 / ESP-NOW

GitHub

- Built an ESP32 wireless head tracking system using ESP-NOW and MPU6500 sensors.
- Designed for VR/AR and robotics with low-latency real-time data transmission.

TOOLS & CONCEPTS

Git/GitHub

Rapid Prototyping

Systems Thinking

API Integration

EDUCATION

B.Tech in ECE

Vidya Jyothi Institute of Technology

2022 - 2026 | CGPA: 8.95/10

Intermediate (MPC)

Sri Bhavishya Junior College

2020 - 2022 | 79.6%

Secondary (SSC)

Dr. K. K. R's Gowtham School

2020 | CGPA: 10/10

LANGUAGES

English (Professional)

Telugu (Native)

Hindi (Conversational)

Experience

Software Developer Intern

Dec 2025 - May 2026

Thoyam Ecocognitives AquaComply Demo

- Developed AquaComply India, an AI-powered environmental compliance tool built with React, Vite, and Tailwind CSS.
- Integrated Google Gemini API and PDF.js to automatically parse lab reports and validate parameters against CPCB standards.

Engineering Intern

Aug 2025 - Nov 2025

Hindustan Aeronautics Limited (HAL)

- Developed a Python-based simulation to model and validate data-link establishment logic between distributed systems.
- Mapped state transitions and failure modes across multi-node communication scenarios.

IoT & Robotics Intern

Mar 2025 - Apr 2025

EduTech Innovations

- Prototyped IoT systems using ESP32, integrating multiple sensors and actuators to accelerate embedded hardware build-test cycles.



Software Portfolio

Scan to view my web apps and software solutions.



Hardware Portfolio

Scan to view my embedded systems and IoT devices.